

Alex Yen

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SKILLS

Languages: *Python, C* | **Tools & Frameworks:** *NumPy, (Geo)Pandas, Matplotlib, Scikit-learn, PyTorch* | **Operating Systems:** *Linux, Windows* | **Other:** *Fusion 360, MQTT, Arduino IDE, Data Structures & Algorithms, XCAP/XCAL*

EDUCATION

University of California San Diego, San Diego, CA 2020–Present
Ph.D. in Computer Science
Advisor: Pat Pannuto

University of California San Diego, San Diego, CA 2020–2023
M.S. in Computer Science
Advisor: Pat Pannuto

University of Massachusetts, Amherst, Amherst, MA 2016–2020
B.S. in Computer Engineering
Honors Thesis Advisor: Jay Taneja
Summa Cum Laude

AWARDS AND HONORS

National Science Foundation Graduate Research Fellowship 2022
UMass Amherst ECE Award of Excellence (3%) 2020
Best Paper Nominee, BuildSys'19 2019
UMass Amherst Commonwealth Honors College: Honors Research Grant 2018

INDUSTRY EXPERIENCE

Research Intern at T-Mobile (Technology Team) 2024

- Reverse engineered competitor wireless products to inform next-gen, Ambient Internet of Things frameworks via spectrum analysis
- Evaluated 5G system behavior across new vendors and wireless bands, and assessed resource allocation, throughput, and standalone versus non-standalone coverage switching

RESEARCH AND TECHNICAL PROJECTS

Geolocating Street Lights with Stereo Cameras at Night 2020-2023

- Implemented a stereo-vision pipeline to backproject and triangulate the geocentric coordinates of street lights in the 3D world from stereo images and the stereo camera's geodetic coordinates

- Performed camera calibration of intrinsic and extrinsic matrices, rectified stereo images, handled frame rotations from the geocentric coordinate system into the local camera frame, and implemented back-projection and triangulation
- Constructed a stereo camera with 3D printed parts and commodity, Android smartphones
- Achieved an accuracy of around 10 m of error in a best-case dataset, but identified issues with camera calibration that threw off results

Functionality, Performance, and Reliability Analysis of the Helium Network 2020-2024

- Revealed and analyzed trends in the Helium Network’s (in)stability and growth, decentralization, and user adoption by inspecting Helium blockchain data with Python
- Designed a cross-country measurement study, comparing and analyzing cellular’s planned infrastructure versus Helium’s ad-hoc LoRa infrastructure by shipping transmitting devices nationwide via FedEx

LoRa Wide-Area Broadcasting via Non-Terrestrial Vehicles 2024-2025

- Deployed LoRa transmitters on unmanned balloons and airplanes, discovering opportunities to utilize existing aerial platforms to broadcast metropolitan-wide wireless coverage at fractions of traditional infrastructure cost (e.g. towers, satellites)
- Automated data collection and post-processing with MQTT, JavaScript, and Python; developed and modified Arduino firmware in C to enable dynamic transmit power control
- Simulated and visualized LoRa-based coverage from airplane-location data (i.e. ADS-B data) over California with Matplotlib and (Geo)Pandas
- Optimized simulation runtime via data batching, parallelization, and memory management

CONFERENCE PUBLICATIONS

1. **Alex Yen**, Alex Bellon, Kaila Rosing, Zesen Zhang, Aaron Schulman, and Pat Pannuto. ““This is a Test of The Emergency Alert System.” Wide-Area Broadcasting With LoRa in the Air”. *In Submission*.
2. **Alex Yen**, Zeal Shah, Ben Ochoa, Pat Pannuto, and Jay Taneja. “EffiSenseSee: Towards Classifying Light Bulb Types and Energy Efficiency with Camera-Based Sensing.” In the 9th ACM International Conference on Systems for Energy-Efficient Built Environments, Cities, and Transportation (BuildSys’22), November 2022.
3. Dhananjay Jagtap*, **Alex Yen***, Huanlei Wu, Aaron Schulman, and Pat Pannuto. 2021. Federated Infrastructure: Usage, Patterns, and Insights from “The People’s Network”. In ACM Internet Measurement Conference (IMC ’21), November 2021.
4. Zeal Shah, **Alex Yen**, Ajey Pandey, and Jay Taneja. “GridInSight: Monitoring Electricity Using Visible Lights.” In the 6th ACM International Conference on Systems for Energy-Efficient Built Environments, Cities, and Transportation (BuildSys’19), November 2019. **Best Paper Nominee**.

WORKSHOP PUBLICATIONS

1. Alex Bellon, **Alex Yen**, and Pat Pannuto. 2023. TagAlong: Free, Wide-Area Data-Muling and Services. In The 24th International Workshop on Mobile Computing Systems and Applications (HotMobile ’23), February 22–23, 2023, Newport Beach, CA, USA.
2. **Alex Yen**, Bryse Flowers, Wenshan Luo, Nitish Nagesh, Peter Tueller, Ryan Kastner, and Pat Pannuto. 2021. A UCSD view on replication and reproducibility for CPS & IoT. In Proceedings of the Workshop on Benchmarking Cyber-Physical Systems and Internet of Things (CPS-IoTBench ’21), May 2021.

POSTERS AND DEMOS

1. Alex Bellon, **Alex Yen**, and Pat Pannuto. 2023. TagAlong: Free, Wide-Area Data-Muling and Services. In The 24th International Workshop on Mobile Computing Systems and Applications (HotMobile '23), February 22–23, 2023, Newport Beach, CA, USA.
2. Alex Bellon, **Alex Yen**, Pat Pannuto. “TagAlong: A Free, Wide-Area Data-Muling Service Built on the AirTag Protocol.” In the 20th ACM Conference on Embedded Networked Sensor Systems (SenSys'22), November, 2022.
3. Zeal Shah, **Alex Yen**, Ajey Pandey, Jay Taneja. “GridInSight: Monitoring Electricity Using Visible Lights.” In the 2nd Annual ACM SIGCAS Conference on Computing and Sustainable Societies (COMPASS'19), July 2019.

SERVICE AND ACTIVITIES

UCSD NSF GRFP Workshop, Lead Organizer	2022, 2023, 2024
ENLACE Program, Mentor	2022
UCSD GradWIC Mentorship Program, Mentor	2022, 2023
UCSE CSE PhD Visit Day, Embedded-Group Lead Organizer	2022, 2023, 2024
Diversity, Equity, and Inclusion PhD Application Review	2021, 2023
UMass Amherst Science Olympiad Mentoring, Head Organizer	2019
HackUMass VI, Lead Hardware Inventory Manager	2018
HackUMass V, Lead Hardware Inventory Manager	2017